Weighing Optimization on a Single-Pan Balance:

From Bilateral Pascal Truncation to Spectral Shielding

Conceptual, Analytical, and Benchtop Guide

Abstract

This guide presents the theory and practical application of optimal weighing designs on single-pan balances (where objects can only be summed on the same pan) when the goal is to determine a single mass of interest (m_1) in the presence of $p = N - 1$ auxiliary masses. We demonstrate why the naive strategy of including m_1 in nearly all measurements fails as the budget grows, and we establish the general constructive principle: bilateral filling of the ends of Pascal's Triangle (light layers with m_1 and heavy layers without m_1). Using the Schur complement, we derive matrix-inversion-free closed-form benchtop formulas and prove the Optimal Ridge Theorem ($k_A = k_B + 1$), whereby marginal Lagrange equilibrium preserves the universal invariant $\bar{s}/\mu_1 = 1/N$ from the Star design up to the Full Factorial design. Furthermore, we derive the exact efficiency equation at half the factorial budget—which unifies the benchtop gain and the asymptotic Gaussian limit ($\sqrt{2}$)—and solve the slicing of partial layers via cyclic group orbits, difference sets (Fano Plane), and the spectral decomposition of circulant matrices.

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Part I: The Physical Problem and Pascal's Principle

1 The Experimental Scenario and the Gap Between Extremes

Imagine N masses on a laboratory bench. You only have a **single-pan balance**: in each measurement, you are only allowed to sum subsets of objects on the same pan (it is not possible to subtract masses on opposing pans).

Only one of these objects is of actual interest to you—the **primary mass** m_1. The other $p = N - 1$ objects $(m_2, m_3, \ldots, m_N)$ act exclusively as **auxiliary masses** (nuisance parameters/extras) that help reduce statistical uncertainty through combined weighings. The practical problem is:

> *Given an arbitrary budget of m weighings, which combinations of objects should we place on the balance pan to estimate m_1 with the smallest possible standard uncertainty (σ_1)?*

Historically, two extremes mark the boundaries of this problem:

1. **The Star Design ($m = N + 1$):** One weighs m_1 alone, then m_1 paired with each auxiliary mass individually (p weighings), and finally performs a single weighing with all p auxiliary masses together without m_1. It is the minimum viable budget, but it offers low precision as N grows.

2. **The Full Factorial Design ($m = 2^N - 1$):** All possible non-empty combinations are tested. It achieves the maximum precision of a combinatorial cycle, but quickly becomes unfeasible (255 weighings for $N = 8$; 1023 for $N = 10$).

The fundamental question is: **how do we construct the optimal experiment for any budget m situated between the Star and the Full Factorial designs?**

2 The Naive Trap and the Physical Tug-of-War

When we have a budget larger than the minimum ($m \gg N + 1$), the immediate intuition of an unsuspecting operator is usually to extrapolate the structure of the Star design:

"If my sole interest is measuring m_1, I will include m_1 in $m - 1$ weighings and leave just 1 weighing without it to close the system!"

2.1 Why is this intuition a disaster for larger budgets?

In the Star design ($m = N + 1$), reserving only 1 weighing without m_1 works because each auxiliary mass accompanies m_1 only once. However, as m grows, m_1 goes onto the pan accompanied by pairs, triplets, and larger groups of auxiliaries (e.g., $y = m_1 + m_2 + m_3$).

To isolate m_1 from these sums, the mathematical model must repeatedly subtract the influence of $m_2, m_3, \ldots, m_N$. If you keep only 1 weighing without m_1, the reference network of the secondary masses becomes loose, and their uncertainty leaks in a cascade into m_1 via covariance. In a test with $N = 8$ and half the factorial weighings ($m = 129$), this naive choice causes the error of m_1 to **nearly double (+95%)** compared to the optimum.

2.2 The Partition Between Direct Signal (Block A) and Anchoring (Block B)

Every optimal design divides the total budget $m = m_A + m_B$ into two blocks with complementary physical functions:

- **Block A (m_A weighings WITH m_1):** Injects direct signal from the mass of interest, but introduces coupling with the auxiliary masses sharing the pan with it.
- **Block B (m_B weighings WITHOUT m_1):** Acts exclusively to calibrate and lock the auxiliary masses to one another, preventing their noise from contaminating m_1.

3 The Universal Principle: Bilateral Pascal Truncation

When investigating which specific combinations form the optimal matrices across different budgets, one discovers that the general solution is governed by the geometry of **Pascal's Triangle** applied to the $p = N - 1$ auxiliary objects.

Denoting by $r \in \{0, 1, \ldots, p\}$ the number of auxiliaries present on the pan in a given weighing (the auxiliary Hamming weight), each complete layer of Pascal's Triangle contains

$\binom{p}{r}$ combinations. The optimal protocol fills Pascal's Triangle **simultaneously from both opposite ends**:

The Bilateral Construction Law in Pascal's Triangle

1. **In Block A (WITH m_1) — Ascend Through the Light Layers ($r = 0, 1, 2, \ldots, k_A$):** To spare m_1 from contamination, first choose combinations with the **smallest number of auxiliaries**: start with m_1 alone ($r = 0$, layer $\binom{p}{0}$), then m_1 with 1 auxiliary ($r = 1$, layer $\binom{p}{1}$), then with pairs ($r = 2$, layer $\binom{p}{2}$), and so on.

2. **In Block B (WITHOUT m_1) — Descend Through the Heavy Layers ($r = p, p-1, \ldots, p-k_B$):** To lock the sum of the auxiliaries with maximum stiffness per measurement, choose combinations with the **largest number of auxiliaries**: start with all p auxiliaries together ($r = p$, layer $\binom{p}{p}$), then almost all together missing only one ($r = p-1$, layer $\binom{p}{p-1}$), missing two ($r = p-2$), and so forth.

As we will prove in Part II (Section 6), because layer $r = 0$ of Block A (m_1 alone) provides pure signal without coupling to the auxiliaries—whereas layer $r = 0$ of Block B would be an empty pan—the optimal balance of complete layers occurs when Block A is **exactly one layer ahead of Block B ($k_A = k_B + 1$)**.

Part II: Gauss-Markov Formalism and Analytical Solution

4 The Partitioned Linear Model and the Schur Complement

We model the experiment using ordinary least squares (Gauss-Markov). Let $y \in \mathbb{R}^m$ be the vector of balance readings and $\beta = [m_1, m_2, \ldots, m_N]^T$ the vector of true masses:

$$y = X\beta + \varepsilon, \quad \mathbb{E}[\varepsilon] = \mathbf{0}, \quad \mathrm{Cov}(\varepsilon) = \sigma^2 I_m \tag{1}$$

where σ is the instrumental uncertainty of the balance and $X \in \{0,1\}^{m \times N}$ is the binary design matrix. Grouping the m_A weighings of Block A above the m_B weighings of Block B, we partition X:

$$X = \begin{pmatrix} \mathbf{1}_{m_A} & X_A \\ \mathbf{0}_{m_B} & X_B \end{pmatrix} \tag{2}$$

where $X_A \in \{0,1\}^{m_A \times p}$ and $X_B \in \{0,1\}^{m_B \times p}$ describe the presence of the $p = N - 1$ auxiliary objects.

The Fisher information matrix (up to σ^{-2}) takes the block form:

$$X^T X = \begin{pmatrix} m_A & \mathbf{s}^T \\ \mathbf{s} & M_{\text{aux}} \end{pmatrix} \tag{3}$$

where:

- $\mathbf{s} = X_A^T \mathbf{1}_{m_A} \in \mathbb{R}^p$: **coupling vector**, whose entries s_i count how many times auxiliary i went onto the pan together with m_1.
- $M_{\text{aux}} = X_A^T X_A + X_B^T X_B \in \mathbb{R}^{p \times p}$: **accumulated information matrix of the auxiliaries** throughout the entire experiment.

By the Partitioned Matrix Inversion Lemma (**Schur Complement**), the variance of the estimator $\hat{m}_1$ is determined by the $(1,1)$ element of $(X^T X)^{-1}$:

$$\text{Var}(\hat{m}_1) = \sigma^2 [(X^T X)^{-1}]_{11} = \frac{\sigma^2}{\Phi} \quad \Longrightarrow \quad \sigma_1 = \frac{\sigma}{\sqrt{\Phi}} \tag{4}$$

where Φ is the **net information about** m_1:

$$\Phi = \underbrace{m_A}_{\text{Gross Signal}} - \underbrace{\mathbf{s}^T M_{\text{aux}}^{-1} \mathbf{s}}_{\text{Contamination Penalty}} \tag{5}$$

5 Closed-Form Combinatorial Solution (Without Matrix Inversion)

When Blocks A and B are composed of complete layers of Pascal's Triangle (layer sets $\mathcal{L}_A$ and $\mathcal{L}_B$), the matrix problem of Equation (5) reduces to elementary combinatorial counting on paper.

5.1 Layer Properties and Analytical Diagonalization

For a complete layer of auxiliary weight $r \in \{0, 1, \ldots, p\}$, we have (with the convention $\binom{n}{k} = 0$ for $k < 0$ or $k > n$):

1. **Total weighings in the layer:** $C(r) = \binom{p}{r}$.
2. **Appearances of each individual auxiliary:** $k(r) = \binom{p-1}{r-1}$.
3. **Co-occurrences of each distinct pair of auxiliaries:** $\lambda(r) = \binom{p-2}{r-2}$.

Summing over all layers chosen in the protocol ($\mathcal{L}_A$ and $\mathcal{L}_B$), we obtain:

$$\bar{s} = \sum_{r \in \mathcal{L}_A} \binom{p-1}{r-1} \quad \Longrightarrow \quad \mathbf{s} = \bar{s}\, \mathbf{1}_p \tag{6}$$

$$k_{\text{tot}} = \sum_{r \in \mathcal{L}_A \cup \mathcal{L}_B} \binom{p-1}{r-1} \quad \text{(diagonal elements of } M_{\text{aux}}) \tag{7}$$

$$\lambda_{\text{tot}} = \sum_{r \in \mathcal{L}_A \cup \mathcal{L}_B} \binom{p-2}{r-2} \quad \text{(off-diagonal elements of } M_{\text{aux}}) \tag{8}$$

By permutation symmetry among the auxiliaries, $M_{\text{aux}} = (k_{\text{tot}} - \lambda_{\text{tot}}) I_p + \lambda_{\text{tot}} J_p$, where $J_p = \mathbf{1}_p \mathbf{1}_p^T$. Thus, the coupling vector $\mathbf{s} = \bar{s}\mathbf{1}_p$ is an exact eigenvector of M_{aux} associated with the **collective stiffness** eigenvalue μ_1:

$$\mu_1 = k_{\text{tot}} + (p-1)\lambda_{\text{tot}} \tag{9}$$

Consequently, the quadratic form $\mathbf{s}^T M_{\text{aux}}^{-1} \mathbf{s}$ collapses into the closed scalar expression:

$$\Phi = m_A - \frac{p \cdot \bar{s}^2}{\mu_1} = m_A - \frac{p \cdot \bar{s}^2}{k_{\text{tot}} + (p-1)\lambda_{\text{tot}}} \tag{10}$$

5.2 Closed-Form Benchtop Formula for Estimating $\hat{m}_1$

Under the same symmetry ($\mathbf{s} = \bar{s}\mathbf{1}_p$ and $M_{\text{aux}}\mathbf{1}_p = \mu_1\mathbf{1}_p$), it is unnecessary to invert matrices to extract the estimated numerical value $\hat{m}_1$ from the raw readings $y = [y_1, \ldots, y_m]^T$. Expanding the first coordinate of $\hat{\beta} = (X^TX)^{-1}X^Ty$, we obtain:

$$\hat{m}_1 = \frac{1}{\Phi}\left[\sum_{j\in\text{Block } A} y_j - \frac{\bar{s}}{\mu_1}\sum_{j=1}^{m} w_{\text{aux},j}\cdot y_j\right] \tag{11}$$

where $w_{\text{aux},j}$ is the **number of auxiliary objects present on the pan in weighing** j. Physically:

1. The first term ($\sum_{j\in A} y_j$) sums all readings containing m_1, accumulating $m_A m_1 + \bar{s}\sum_{i=2}^{N} m_i$.

2. The second term weights each reading of the experiment by the number of auxiliaries on the pan, totaling $\mu_1\sum_{i=2}^{N} m_i + p\bar{s}m_1$. Multiplied by $\bar{s}/\mu_1$, it **identically cancels the sum of the auxiliary masses**.

3. The factor $1/\Phi$ normalizes the remaining net signal, yielding the exact $\hat{m}_1$ using only scalar sums and multiplications at the bench.

6 Lagrange Optimality and the Ridge Theorem $k_A = k_B + 1$

When filling Block A with light layers $r \in \{0, 1, \ldots, k_A\}$ and Block B with heavy layers $r \in \{p, p-1, \ldots, p-k_B\}$, **what is the optimal relationship between the depths k_A and k_B, and when should it be supplemented by orbital splits?**

6.1 Marginal Derivatives and the Lagrange Equilibrium Condition

Consider any symmetric state parameterized by $(m_A, \bar{s}, \mu_1)$. If we add a symmetric package of Δm weighings of constant auxiliary weight:

- **Addition in Block A (auxiliary weight r_A):** We have $\Delta m_A = \Delta m$, $\Delta\bar{s} = \Delta m\frac{r_A}{p}$, and $\Delta\mu_1 = \Delta m\frac{r_A^2}{p}$. Substituting into the finite variation of $\Phi = m_A - p\bar{s}^2/\mu_1$, the quadratic terms in Δm^2 in the numerator cancel out identically, resulting in:

$$\Delta\Phi_A(r_A, \Delta m) = \frac{\Delta m\left(1 - r_A\frac{\bar{s}}{\mu_1}\right)^2}{1 + \frac{\Delta m\, r_A^2}{p\mu_1}} \xrightarrow{\Delta m\to 0} \left.\frac{\partial\Phi}{\partial m_A}\right|_{r_A} = \left(1 - r_A\frac{\bar{s}}{\mu_1}\right)^2 \tag{12}$$

- **Addition in Block B (auxiliary weight r_B):** We have $\Delta m_A = 0$, $\Delta\bar{s} = 0$, and $\Delta\mu_1 = \Delta m\frac{r_B^2}{p}$, resulting in:

$$\Delta\Phi_B(r_B, \Delta m) = \frac{\Delta m\left(r_B\frac{\bar{s}}{\mu_1}\right)^2}{1 + \frac{\Delta m\, r_B^2}{p\mu_1}} \xrightarrow{\Delta m\to 0} \left.\frac{\partial\Phi}{\partial m_B}\right|_{r_B} = \left(r_B\frac{\bar{s}}{\mu_1}\right)^2 \tag{13}$$

By the method of Lagrange multipliers, the optimal budget partition occurs at the stationary boundary where the marginal gain of the next available layer in A (r_A) equals the marginal gain of the next available layer in B (r_B):

$$\left.\frac{\partial \Phi}{\partial m_A}\right|_{r_A} = \left.\frac{\partial \Phi}{\partial m_B}\right|_{r_B} \iff 1 - r_A \frac{\bar{s}}{\mu_1} = r_B \frac{\bar{s}}{\mu_1} \iff \frac{\bar{\mathbf{s}}}{\boldsymbol{\mu_1}} = \frac{\mathbf{1}}{\mathbf{r_A} + \mathbf{r_B}} \tag{14}$$

6.2 The Algebraic Identity $\mu_1 = N\bar{s}$ on the Diagonal $k_A = k_B + 1$

By the combinatorial identity $\binom{p-1}{r-1} + (p-1)\binom{p-2}{r-2} = r\binom{p-1}{r-1}$, each complete layer of auxiliary weight r contributes exactly $r\binom{p-1}{r-1}$ to the eigenvalue μ_1. Making the index change $u = r - 1$ in Block A and $u = p - r$ in Block B (using $\binom{p-1}{p-u-1} = \binom{p-1}{u}$), we obtain for any pair (k_A, k_B):

$$\bar{s}(k_A) = \sum_{u=0}^{k_A-1} \binom{p-1}{u}, \qquad \mu_1(k_A, k_B) = \sum_{u=0}^{k_A-1} (u+1)\binom{p-1}{u} + \sum_{u=0}^{k_B} (p-u)\binom{p-1}{u} \tag{15}$$

When we impose $k_A = k+1$ **and** $k_B = k$ (that is, $k_A = k_B + 1$), the upper limits of the two sums in μ_1 become identical ($u = 0, \ldots, k$), allowing us to merge them term by term:

$$\mu_1(k+1, k) = \sum_{u=0}^{k} \underbrace{[(u+1) + (p-u)]}_{=p+1=N} \binom{p-1}{u} = (p+1)\bar{s}(k+1) = N\bar{s}(k+1) \tag{16}$$

Therefore, across the entire family $k_A = k_B + 1$ (for any N and any depth $k \in \{0, \ldots, p-1\}$), the **exact spectral invariant** holds:

$$\frac{\bar{s}}{\mu_1} \equiv \frac{1}{p+1} = \frac{1}{N} \tag{17}$$

Furthermore, starting from the state $(k_A = k+1, k_B = k)$, the next boundary layers are $r_A = k+2$ and $r_B = p-k-1$, whose sum is exactly $r_A + r_B = p+1 = N$. Substituting into Equation (14), we conclude that **the diagonal $k_A = k_B + 1$ exactly satisfies the stationary Lagrange condition**:

$$\left.\frac{\partial \Phi}{\partial m_A}\right|_{r_A=k+2} = \left.\frac{\partial \Phi}{\partial m_B}\right|_{r_B=p-k-1} = \left(\frac{p-k-1}{p+1}\right)^2 \tag{18}$$

Universal Simplification on the Optimal Ridge ($\bar{s}/\mu_1 = 1/N$)

In any symmetric configuration that preserves the Lagrange invariant $\bar{s}/\mu_1 = 1/N$ (including all Clean Points $k_A = k_B + 1$, from the Star Design $k = 0$ to the Full Factorial $k = p - 1$), Equations (10) and (11) take their simplest possible form:

$$\Phi = m_A - \frac{N-1}{N}\bar{s}, \qquad \hat{m}_1 = \frac{1}{\Phi}\left[\sum_{j \in \text{Block } A} y_j - \frac{1}{N}\sum_{j=1}^{m} w_{\text{aux},j} \cdot y_j\right] \tag{19}$$

6.3 Off-Ridge Analysis and the Central Layer Parabola

Equations (12) and (13) determine the system's behavior in three fundamental scenarios:

1. **Strict Dominance Under the Same Budget (Odd N):** For $N = 7$ ($p = 6$) and budget $m = 64$, the symmetry $\binom{6}{4} = \binom{6}{2} = 15$ allows forming three complete-layer configurations with exactly 64 weighings:

   - $(k_A = 2, k_B = 3) \implies k_A = k_B - 1$ $(m_A = 22, m_B = 42)$: $\Phi = 22 - \frac{6\times 6^2}{112} =$ **20.071**.
   - $(k_A = 4, k_B = 1) \implies k_A = k_B + 3$ $(m_A = 57, m_B = 7)$: $\Phi = 57 - \frac{6\times 26^2}{112} =$ **20.786**.
   - $(k_A = 3, k_B = 2) \implies k_A = k_B + 1$ $(m_A = 42, m_B = 22)$: $\Phi = 42 - \frac{6\times 16^2}{112} =$ **28.286**.

   With the same 64 weighings, the ridge $k_A = k_B + 1$ delivers +36.1% more net information than $(4, 1)$ and +40.9% more than $(2, 3)$.

2. **The Central Layer Parabola ($m = 2^{N-1}$ for Even N / Odd p):** When $p = 2q + 1$ is odd (e.g., $N = 8, p = 7, q = 3$), starting from the Clean Point $(q, q - 1) = (3, 2)$ $(m = 93)$, the next boundary layers in A and B coincide at the **same central layer** $r_A = r_B = q + 1 = 4$, of size $C = \binom{7}{4} = 35$. To reach half the factorial budget $(m = 93 + 35 = 128)$, if we allocate Δm_A weighings of weight $r = 4$ to Block A and $\Delta m_B = C - \Delta m_A$ weighings of weight $r = 4$ to Block B, the denominator $\mu_1 = (p+1)2^{p-2} = 256$ remains constant (as it depends only on the sum $\Delta m_A + \Delta m_B = C$), and the net information traces an **exact concave parabola**:

$$\Phi(\Delta m_A) = \Phi_{\text{ends}} + \frac{(q+1)^2}{p(p+1)2^{p-2}}\,\Delta m_A(C - \Delta m_A) \tag{20}$$

   For $N = 8$ $(m = 128)$, Equation (20) yields:

$$\Phi(\Delta m_A) = 50.765625 + \frac{\Delta m_A(35 - \Delta m_A)}{112} \tag{21}$$

   Therefore, dumping the entire $r = 4$ layer into B—generating $(3, 3)$ with $\Delta m_A = 0$—or entirely into A—generating $(4, 2)$ with $\Delta m_A = 35$—produces the two **minima** of the parabola ($\Phi = 50.77$, error of +19.1%), as it drives $\bar{s}/\mu_1$ away from $1/8$. Conversely, **splitting the central layer equally between A and B** ($\Delta m_A = \Delta m_B = C/2 = 17.5$) exactly restores the Lagrange invariant $\bar{s}/\mu_1 = 1/N = 1/8$ and achieves the **global maximum**:

   - **Division into complete cyclic orbits of size 7 (3 orbits in A and 2 in B, $\Delta m_A = 21, \Delta m_B = 14$, partition $85/43 = 66.4\%$):** $\Phi = 50.765625 + \frac{21\times 14}{112} =$ **53.3906** (+16.1% error, retaining a closed-form manual solution).
   - **Maximum integer split ($\Delta m_A = 18, \Delta m_B = 17$, partition $82/46 = 64.1\%$):** $\Phi =$ **53.4978** (+16.0% error; with the 129th weighing at $m = 129$, it reaches $\Phi =$ **53.756** and +15.7% error).

3. **Transition Between Two Consecutive $k_A = k_B + 1$ Points:** As shown in Table 1 for $N = 8$ between $(2, 1)$ ($m = 37$) and $(3, 2)$ ($m = 93$), advancing two consecutive full layers in A—point $(3, 1)$ at $m = 72$ ($\Phi = 30.455$)—severely degrades the marginal return due to excess coupling ($\bar{s}/\mu_1 = 0.200 \gg 1/8$), losing even to $m = 58$ with interleaved orbits (2 orbits in A + 1 orbit in B, which keeps $\bar{s}/\mu_1 = 13/99 \approx 1/8$ and achieves $\Phi = \mathbf{31.051}$).

Table 1: Comparison of Configurations Between $m = 37$ and $m = 128$ for $N = 8$ ($p = 7$)

Configuration ($N = 8$)	**Structure**	m	m_A/m_B	$\bar{s}/\mu_1$	**Information** (Φ)	σ_1/σ
Clean Point $(2, 1)$	$k_A = k_B + 1$	37	29/8	0.1250 (= 1/8)	22.875	0.2091
Full Layer $(2, 2)$	$k_A = k_B$	58	29/29	0.0519	26.382	0.1947
Base $(2, 1)$ + **Orbits** ($2A + 1B$)	**Interleaved**	**58**	**43/15**	**0.1313** ($\approx 1/8$)	**31.051**	**0.1795**
Base $(2, 2)$ + 1 Orbit in A	Fractional	65	36/29	0.0714	31.000	0.1796
Full Layer $(3, 1)$	$k_A = k_B + 2$	72	64/8	0.2000	30.455	0.1812
Clean Point $(3, 2)$	$k_A = k_B + 1$	93	64/29	0.1250 (= 1/8)	44.750	0.1495
Full Layers $(3, 3)$ or $(4, 2)$	Ends $r = 4$	128	64/64 or 99/29	0.0859 or 0.1641	50.766	0.1403
Base $(3, 2)$ + **Orbits** ($3A + 2B$)	**Orbital Ridge**	**128**	**85/43**	**0.1328** ($\approx 1/8$)	**53.391**	**0.1369**
Base $(3, 2)$ + **Central Split** ($18A + 17B$)	**Parabola Vertex**	**128**	**82/46**	**0.1261** ($\approx 1/8$)	**53.498**	**0.1367**

Part III: Budget Engineering and Asymptotic Limit

7 Clean Points, Unified Half-Budget Equation, and Gaussian Limit

When there is freedom to choose the number of weighings m at the bench, the simplest choice is to adopt a **Clean Point** on the ridge $k_A = k_B + 1$ (Table 2), where complete binomial layers natively guarantee $\bar{s}/\mu_1 = 1/N$.

7.1 The Exact Analytical Law at the 50% Cut ($m = 2^{N-1}$) and the $\sqrt{2}$ Limit

To compare the Pascal design at half the budget ($m = 2^p = 2^{N-1}$) with the full factorial ($m = 2^N - 1$, where $m_A = 2^p$, $\bar{s} = 2^{p-1}$, and $\Phi_{\text{full}} = 2^{p-1}\frac{p+2}{p+1}$), we keep the system on the

Table 2: Complete-Layer Clean Points on the Ridge $k_A = k_B + 1$ by Number of Objects (N)

Objects (N)	**Auxiliaries** (p)	**Clean Budget** (m)	**Partition** (m_A/m_B)	**Layers Used** (r)	**Fraction in** A
4	3	**11**	7 / 4	A: $\{0,1,2\}$; B: $\{3,2\}$	63.6%
5	4	**16**	11 / 5	A: $\{0,1,2\}$; B: $\{4,3\}$	68.8%
6	5	**22**	16 / 6	A: $\{0,1,2\}$; B: $\{5,4\}$	72.7%
7	6	**64**	42 / 22	A: $\{0,1,2,3\}$; B: $\{6,5,4\}$	65.6%
8	7	**93**	64 / 29	A: $\{0,1,2,3\}$; B: $\{7,6,5\}$	68.8%
8	7	**163**	99 / 64	A: $\{0,1,2,3,4\}$; B: $\{7,6,5,4\}$	60.7%

Lagrange ridge $\bar{s}/\mu_1 = 1/N$ (which, for odd N, corresponds to the Clean Point $(q, q-1)$ and, for even N, corresponds to the vertex of the parabola in Equation 20 by splitting the central layer equally between A and B).

In both cases (even or odd N), the optimal parameters at $m = 2^p = 2^{N-1}$ obey the exact identities:

$$m_A = 2^{p-1} + \frac{1}{2}\binom{p}{\lfloor p/2 \rfloor}, \qquad m_B = 2^{p-1} - \frac{1}{2}\binom{p}{\lfloor p/2 \rfloor} \quad \Longrightarrow \quad \frac{\mathbf{m_A}}{\mathbf{m}} = \frac{\mathbf{1}}{\mathbf{2}} + \frac{\binom{\mathbf{p}}{\lfloor \mathbf{p/2} \rfloor}}{\mathbf{2^{p+1}}} \tag{22}$$

and $\bar{s} = 2^{p-2}$, $\mu_1 = (p+1)2^{p-2}$, which yields the **unified formula for net information at half the budget**:

$$\Phi_{\text{Pascal}}(2^{N-1}) = \frac{1}{2}\Phi_{\text{full}} + \frac{1}{2}\binom{p}{\lfloor p/2 \rfloor} \quad \Longrightarrow \quad \frac{\mathbf{\Phi_{\text{Pascal}}(2^{N-1})}}{\mathbf{\Phi_{\text{full}}}} = \frac{\mathbf{1}}{\mathbf{2}} + \frac{\mathbf{p+1}}{\mathbf{p+2}} \cdot \frac{\binom{\mathbf{p}}{\lfloor \mathbf{p/2} \rfloor}}{\mathbf{2^p}} \tag{23}$$

Equations (22) and (23) simultaneously govern the benchtop regime and the asymptotic limit:

1. **In the Practical Benchtop Regime ($N \leq 10$):** The relative weight of the central binomial layer $\binom{p}{\lfloor p/2 \rfloor}/2^p$ is high (between 0.27 and 0.38 for $N \in [4, 8]$). By Equation (22), this shifts the optimal fraction of appearances of m_1 to **64**% to **69**% ($\approx 2/3$ of the budget) and raises the ratio $\Phi_{\text{Pascal}}/\Phi_{\text{full}}$ to ≈ 0.74 to 0.81 ($\approx 3/4$), keeping the uncertainty increase restricted to **+10**% to **+16**% (Table 3).

2. **In the Asymptotic Gaussian Limit ($N \to \infty$):** By Stirling's approximation (De Moivre-Laplace Central Limit Theorem), the density of the central layer decays as:

$$\frac{\binom{p}{\lfloor p/2 \rfloor}}{2^p} \sim \sqrt{\frac{2}{\pi p}} \xrightarrow{p \to \infty} 0 \tag{24}$$

   Consequently, as $N \to \infty$, the optimal fraction converges to $\lim_{N\to\infty}(m_A/m) = \mathbf{50}\%$, the net information converges to $\lim_{N\to\infty}(\Phi_{\text{Pascal}}/\Phi_{\text{full}}) = \mathbf{1/2}$ (exact doubling of the variance when cutting half the observations), and the uncertainty ratio converges to the **Asymptotic Gaussian Limit**:

$$\lim_{N\to\infty} \frac{\sigma_1(\text{Pascal})}{\sigma_1(\text{full})} = \sqrt{2} \approx \mathbf{1.4142} \quad (+41.4\% \text{ uncertainty}) \tag{25}$$

Table 3: Performance at the 50% Cut of the Full Factorial (from $N = 3$ to the Asymptotic Limit $N \to \infty$)

N	**Factorial** $(2^N - 1)$	**Half Budg.** (m)	**Fraction** m_A/m	$\Phi_{\mathbf{Pascal}}/\Phi_{\mathbf{full}}$	**Increase at** 2^{N-1}	**Increase at** $2^{N-1}+1$
3	7	4 or 5	75.0%	0.8750	+6.9%	+5.4%
4	15	8 or 9	68.8%	0.8000	+11.8%	+8.5%
5	31	16 or 17	68.8%	0.8125	+10.9%	+10.1%
6	63	32 or 33	65.6%	0.7679	+14.1%	+13.3%
7	127	64 or 65	65.6%	0.7734	+13.7%	+13.4%
8	255	128 or 129	63.7%	0.7431	**+16.0%**	**+15.7%**
20	10^6	5.2×10^5	58.8%	0.6673	+22.4%	+22.4%
100	10^{30}	5×10^{29}	54.0%	0.5788	+31.4%	+31.4%
∞	∞	∞	**50.0%**	**0.5000**	**+41.4%** ($\sqrt{2}$)	**+41.4%** ($\sqrt{2}$)

8 Layer Slicing and Spectral Shielding

When the budget does not coincide with a complete-layer Clean Point and forces the selection of only a fraction of a layer $\binom{p}{r}$ (as in the intermediate orbits of Table 1), the way these partial combinations are chosen is decisive for the precision of $\hat{m}_1$.

8.1 The Pitfall of Standard Lexicographical Order

Consider $N = 8$ ($p = 7$ auxiliaries, $m_2, \ldots, m_8$) with budget $m = 65$: complete layers $A_0 = \{0, 1, 2\}$ (29 weighings) and $B_0 = \{7, 6, 5\}$ (29 weighings), plus 7 **extra weighings** drawn from the $\binom{7}{3} = 35$ triplets of layer $r = 3$ of Block A.

If the operator selects the first 7 triplets generated in **lexicographical order** (such as `itertools.combinations` in Python):

$$\begin{array}{ll} \text{Extra weighings 1 to 5:} & m_1 + \mathbf{m_2} + \mathbf{m_3} + m_k \quad (k \in \{4, 5, 6, 7, 8\}) \\ \text{Extra weighings 6 and 7:} & m_1 + \mathbf{m_2} + m_4 + m_k \quad (k \in \{5, 6\}) \end{array}$$

mass m_2 appears in **all 7 weighings** (100%), m_3 appears **5 times**, and m_7, m_8 appear **only once**. This asymmetry distorts the vector $\mathbf{s}$ and projects the error of the undersampled masses onto $\hat{m}_1$.

8.2 The Optimal Solution: Cyclic Orbits and the Fano Plane

To slice a layer $\binom{p}{r}$ while preserving symmetry in packages that are multiples of p, we generate the combinations via **orbits of the cyclic group** C_p (modular rotation over $\{0, 1, \ldots, p-1\}$):

$$\mathcal{S}_k = \pi^k(\mathcal{S}_0) = \{(c_1 + k) \bmod p, \ldots, (c_r + k) \bmod p\}, \quad k = 0, 1, \ldots, p-1 \tag{26}$$

Any seed $\mathcal{S}_0$ guarantees that each auxiliary appears exactly r times in the orbit (**1st-Order Isotropy**: $\mathbf{s} = \bar{s}\mathbf{1}_p$). Additionally, choosing a seed with uniformly distributed modular differences guarantees **2nd-Order Isotropy**:

- **Consecutive Seed** $\{0, 1, 2\}$ (mod 7)**:** The modular differences cover ± 1 and ± 2, but **omit** ± 3 ($\lambda_{ij} \in \{0, 2\}$).

- **Spaced Seed** $\{0,1,3\}$ (mod 7) **(Perfect Difference Set):** The differences cover all residues $\{1,2,3,4,5,6\}$ exactly once, generating the **Fano Projective Plane** 2-$(7,3,1)$: each auxiliary appears $k_i = 3$ times and **every pair shares the panexactly** $\lambda_{ij} = 1$ **time** (Table 4).

Table 4: Benchtop Protocol: The 7 Extra Weighings Generated by the Fano Orbit $\{0,1,3\}$ (mod 7)

Extra Weighing	**Primary Mass**	**Auxiliaries on Pan**	**Modular Indices** ($p = 7$)
1	m_1	$+\,m_2 + m_3 + m_5$	$\{0,1,3\}$ (Seed)
2	m_1	$+\,m_3 + m_4 + m_6$	$\{1,2,4\}$ (+1 mod 7)
3	m_1	$+\,m_4 + m_5 + m_7$	$\{2,3,5\}$ (+2 mod 7)
4	m_1	$+\,m_5 + m_6 + m_8$	$\{3,4,6\}$ (+3 mod 7)
5	m_1	$+\,m_6 + m_7 + m_2$	$\{4,5,0\}$ (+4 mod 7)
6	m_1	$+\,m_7 + m_8 + m_3$	$\{5,6,1\}$ (+5 mod 7)
7	m_1	$+\,m_8 + m_2 + m_4$	$\{6,0,2\}$ (+6 mod 7)

8.3 Proof via Circulant Matrices and Spectral Cancellation

Because the orbit is generated by circular shifts, its information matrix $M_{\text{orb}} = X_{\text{orb}}^T X_{\text{orb}}$ is **symmetric circulant**. Added to the matrix M_{base} of the complete layers, the global matrix $M_{\text{aux}} = M_{\text{base}} + M_{\text{orb}}$ preserves constant row sums, ensuring that $\mathbf{v}_1 = \frac{1}{\sqrt{p}}\mathbf{1}_p$ remains an **exact eigenvector of** M_{aux}.

Expanding the Schur penalty in the orthonormal eigenvectors $\{\mathbf{v}_1, \ldots, \mathbf{v}_p\}$ of M_{aux}:

$$\mathcal{P} = \mathbf{s}^T M_{\text{aux}}^{-1} \mathbf{s} = \underbrace{\frac{(\mathbf{v}_1^T \mathbf{s})^2}{\mu_1}}_{\text{Collective Mode}} + \underbrace{\sum_{j=2}^{p} \frac{(\mathbf{v}_j^T \mathbf{s})^2}{\mu_j}}_{\text{Parasitic Contrast Leakage}} \tag{27}$$

Since $\mathbf{v}_j^T \mathbf{1}_p = 0$ for every contrast mode ($j \geq 2$) and the orbit maintains $\mathbf{s} = \bar{s}\mathbf{1}_p$, we have $\mathbf{v}_j^T \mathbf{s} \equiv 0$. The parasitic leakage vanishes identically (Table 5) and the closed-form formulas of Section 5 remain exact.

Table 5: Spectral Decomposition of the Schur Penalty ($N = 8, p = 7, m = 65$, selection of 7 triplets in $\binom{7}{3}$)

Selection Strategy for the 7 Triplets	σ_1/σ	**Total Penalty**	**Collective Mode**	**Parasitic Leakage**
Cyclic Orbit $\{0,1,3\}$	**0.17961**	**5.00000**	5.00000	**0.00000**
Lexicographical Order (Python)*	0.18300	6.14022	5.08029	**+1.05993**
Worst Case (Maximum Asymmetry)*	0.18300	6.14022	5.08029	**+1.05993**

***Note:** The lexicographical order coincides exactly with the worst possible case within the layer because it freezes m_2 and m_3 at the maximum frequency allowed for $k = 7$ subsets.

8.4 Sub-Orbital Budgets ($k < p$): Monotonicity vs. Manual Solvability

If only $1 \leq k < p$ stray weighings remain (insufficient to close an orbit of size p), the Loewner order ($M_{m+1} = X^T X + xx^T \succeq X^T X$) and the Sherman-Morrison identity guarantee that:

$$\mathrm{Var}(\hat{m}_1)_{m+1} = \mathrm{Var}(\hat{m}_1)_m - \sigma^2 \frac{\left(e_1^T (X^T X)^{-1} x\right)^2}{1 + x^T (X^T X)^{-1} x} \leq \mathrm{Var}(\hat{m}_1)_m \tag{28}$$

Thus, **adding a valid weighing never increases the absolute uncertainty** σ_1. However, interrupting an orbit ($k < p$) reduces the marginal gain per measurement ($\Delta\Phi/\Delta m$) and breaks the condition $\mathbf{s} = \bar{s}\mathbf{1}_p$, requiring a computer to solve $(X^T X)\hat{\beta} = X^T y$.

Part IV: Summary Benchtop Protocol

9 Decision Rules for the Experimentalist

1. **Sizing on the Optimal Ridge ($k_A = k_B + 1$):** Whenever possible, choose a Clean Point where Block A (with m_1) contains the light layers $r = 0, 1, \ldots, k+1$ and Block B (without m_1) contains the heavy layers $r = p, p-1, \ldots, p-k$ (Table 2). At these points, $\bar{s}/\mu_1 = 1/N$ is invariant and the benchtop calculation reduces to:

$$\Phi = m_A - \frac{N-1}{N}\bar{s}, \qquad \hat{m}_1 = \frac{1}{\Phi}\left[\sum_{j \in \text{Block } A} y_j - \frac{1}{N}\sum_{j=1}^{m} w_{\text{aux},j} \cdot y_j\right], \qquad \sigma_1 = \frac{\sigma}{\sqrt{\Phi}}$$

2. **Transition Between Clean Points and Central Layer (Cyclic Orbits in Multiples of p):** For intermediate budgets between two Clean Points $(k+1, k)$ and $(k+2, k+1)$—including $m = 2^{N-1}$ for even N—add packages of p weighings generated by circular rotation $\bmod p$ with a spaced-difference seed (such as $\{0, 1, 3\} \pmod 7$), dividing the orbits between A and B to keep $\bar{s}/\mu_1 \approx 1/N$ and applying the general closed-form formula of Section 5. Never select combinations in alphabetical/lexicographical order.

3. **Sub-Orbital Residuals ($k < p$):** If the calculation is performed manually at the bench, stop at the last complete orbit or Clean Point. If a computer is available, include the k extra weighings by choosing the orbit steps that minimize the frequency dispersion $\mathrm{Var}(s_i)$.